Evaluation Metrics Used

Hadith Hallucination Benchmark — Bilingual Recall Evaluation of Open-Weight LLMs

This note summarizes the metrics used to evaluate whether local LLMs can recall exact Hadith text given only a bibliographic reference (book, chapter, Hadith number), in English and Arabic, and why each metric was chosen.

1. BLEU — Lexical Fidelity

Sentence-level BLEU-4 (uniform n-gram weights, with smoothing) computed between the model's response and the exact ground-truth Hadith text, in the response's own language. Hadith recall is a *verbatim-recall* task: for a given reference there is exactly one correct wording, so a strict n-gram overlap metric — borrowed from machine translation evaluation — is a legitimate proxy for whether the model actually remembered the text, rather than merely produced something topically plausible. BLEU scores near 1.0 for verbatim recall and near 0 for paraphrase or fabrication.

2. Cosine Similarity — Semantic Fidelity

Cosine similarity between multilingual sentence embeddings (`paraphrase-multilingual-MiniLM-L12-v2`) of the model's response and the ground truth. This captures partial credit that BLEU cannot: whether an incorrect response is at least topically related to the correct Hadith (e.g., invoking a plausible but wrong Hadith on a similar theme) versus being entirely unrelated content.

3. Refusal Detection — Rule-Based Filter

Not a similarity metric, but a filtering step applied before scoring: a phrase-match list combined with a response-length cutoff (≤150 characters) used to separate genuine refusals (the model declines to answer or asks for the reference again) from responses that briefly hedge and then still fabricate a full Hadith. This distinguishes an honest 'I don't know' from a confident fabrication, which is an important safety distinction that a similarity score alone cannot make.

4. Five-Way Classification — Composite Outcome

BLEU and cosine similarity are combined via fixed thresholds (τ_{text}=0.4 on BLEU; τ_{low}=0.5, τ_{high}=0.8 on cosine) into a single reported label per response: **correct** (passes both thresholds), **theme_only** (on-theme, wrong wording), **partial** (some overlap on one axis only), **hallucinated** (unrelated to ground truth), or **refused**. This composite label is the primary outcome variable reported in the results tables.

Metrics Considered and Rejected

The following alternatives were evaluated during development and explicitly not used in the final pipeline, for the reasons noted:

- **BERTScore:** Tried as an alternative to BLEU. Its baseline similarity was too high on both English and Arabic text, so it failed to discriminate fabricated responses from genuinely correct ones — reverted to BLEU, which has a naturally low baseline for unrelated text.
- **LLM-as-a-judge:** A separate LLM (Mistral) was tried as a grading judge. Removed because it introduced an uncontrolled second model into the evaluation loop, making results harder to reproduce and justify.
- **Cross-encoder re-ranker:** Explored as an alternative semantic-match signal to cosine similarity. Not adopted in the final pipeline.

Why BLEU (Not Usually the Default Choice)

BLEU is often considered too rigid for evaluating open-ended LLM generation, since natural responses can be correct while phrased differently. That critique does not apply here: Hadith recall is closed-form and memorization-required — the model is being asked to reproduce a fixed, previously-published text, not to generate a novel but semantically equivalent answer. Under that framing, strict lexical overlap is the appropriate tool rather than a limitation.

Summary Table

Metric	Type	Scope	Role
BLEU-4	Lexical	All responses, both languages	Primary correctness signal
Cosine similarity	Semantic	All responses, both languages	Partial-credit / theme signal
Refusal filter	Rule-based	Applied before scoring	Separates refusal from fabrication
5-way label	Composite	Derived from BLEU + cosine	Reported outcome variable